

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		Intron {does not code / spliced out/ not translated} (1) Exon {codes/ is translated} (for a polypeptide) (1) NOT codes for an amino acid(s)	2			2		
		(ii)		Water potential falls (in the mitochondria)/ solute potential decreases (1) Water enters by <u>osmosis</u> (1) Max 1 if refer to cell		2		2		
	(b)	(i)		Phenotype parents: unaffected male, {unaffected/carrier} female (1) Genotype parents, $X^D Y$, $X^D X^d$ (1) Genotype gametes, X^D , Y , X^d , X^d (1) ECF Genotypes offspring, $X^D Y$, $X^d Y$, $X^D X^d$, $X^d X^d$ (1) ECF		4		4		
		(ii)		They could {have two copies of the healthy gene/ be homozygous dominant} or be {heterozygous / carrier}/ They could be $X^D X^d$ or $X^D X^D$ / can't determine until they reproduce (1)		1		1		
	(c)			Any 2 x (1) from: Cause immune response against virus/ antibodies may be produced against it (1) Problems introducing gene into muscle (cells)/ may not reach {target (cell)/ muscle (cell)}/ may invade {non target / host} cells(1) Virus may {become pathogenic/ cause disease/ cause infection/ destroy cells}(1) NOT harm/ illness May affect other genes/ reference to oncogenes(1)	2			2		
	(d)	(i)		CCGUUA(1)	1			1		